

Assignment 1: Symmetric-Key Encryption

ECE 407 – Cryptography · Fall 2026

Covers Weeks 1–4 · 100 points · Due Thursday 09/24/2026, 11:59 PM

Instructions. Answer all four questions and justify each step. Submit one PDF compiled from L^AT_EX; no Word, Google Docs, or photographs of handwritten work. No late submissions. You may use the lecture notes and Katz–Lindell (3rd ed.), and you may discuss the problems with classmates, but write your solutions alone and list your collaborators at the top of your PDF. AI tools are not allowed on this assignment. Logarithms are base 2, n is the security parameter, and “efficient” means probabilistic polynomial time (PPT).

For a security proof, give the reduction: the distinguisher or adversary you construct, what it does with its own input, how it simulates the view of the one it calls, and an equation relating the two advantages. For an attack, give the algorithm, compute or bound its advantage, and check that it runs in polynomial time. A yes-or-no answer with no argument earns almost no credit.

The problems are set in a machine-learning serving system, but the setting is only there to fix the parameters; no background beyond Weeks 1–4 is needed.

Problem 1. Perfect secrecy (25 pts)

An inference service encrypts a short routing code that says which model tier should serve a request. The code is decrypted inside the trust boundary, but the proxy that forwards and logs the request must not learn the tier. A fresh key is used each time.

- (a) **(10 pts)** There are six tiers, so $\mathcal{M} = \{0, 1, 2, 3, 4, 5\}$. A proposed scheme: **Gen** outputs a uniform $k \in \{0, \dots, 7\}$, $\text{Enc}_k(m) = [k + m \bmod 6]$, and $\text{Dec}_k(c) = [c - k \bmod 6]$. Verify that the scheme is correct, then decide whether it is perfectly secret and prove your answer. The key space here is larger than the message space; explain why that alone does not settle the question.
- (b) **(10 pts)** A second service encrypts n -bit strings ($n \geq 2$) carrying a parity check in the last bit:

$$\mathcal{M} = \left\{ m \in \{0, 1\}^n : \bigoplus_{i=1}^n m_i = 0 \right\}.$$

Gen outputs a uniform $k \in \{0, 1\}^{n-1}$. Writing $\pi(k) = \bigoplus_{i=1}^{n-1} k_i$, the scheme sets $\text{Enc}_k(m) = m \oplus (k \parallel \pi(k))$ and $\text{Dec}_k(c) = c \oplus (k \parallel \pi(k))$. Decide whether this scheme is perfectly secret over $\mathcal{M}$ and prove your answer. State the ciphertext space explicitly; a proof that does not identify $\mathcal{C}$ is not complete.

- (c) **(5 pts)** Consider the claim: Π is perfectly secret if and only if $\Pr[\text{Enc}_K(m) = c] = 1/|\mathcal{C}|$ for every $m \in \mathcal{M}$ and every $c \in \mathcal{C}$, where the probability is over $K \leftarrow \text{Gen}$. Prove or refute each direction separately. For a direction you refute, give a counterexample with **Gen**, **Enc**, **Dec**, $\mathcal{M}$, $\mathcal{K}$, and $\mathcal{C}$ all specified, and state a corrected version of the claim.

Problem 2. Key length (20 pts)

A vendor stores user prompts encrypted and claims two things: each prompt is encrypted under a 256-bit key, and the encryption is secure against an attacker with unlimited computing power. Check whether both claims can hold at once.

- (a) **(12 pts)** Prove: if an encryption scheme is perfectly secret, then it has at least as many keys as messages, $|\mathcal{K}| \geq |\mathcal{M}|$. Assume Dec is deterministic and the scheme is correct. You may follow this outline or give your own argument.
1. Assume the opposite, $|\mathcal{K}| < |\mathcal{M}|$, and let the message be uniform over $\mathcal{M}$.
 2. Pick a ciphertext c with $\Pr[C = c] > 0$. Why must there be a message m with $\Pr[\text{Enc}_K(m) = c] > 0$?
 3. Consider the set of messages that c could decrypt to, $\mathcal{M}(c) = \{\text{Dec}_k(c) : k \in \mathcal{K}\}$. How large can it be, and why?
 4. Conclude that some m' lies outside this set. What is $\Pr[\text{Enc}_K(m') = c]$, and why?
 5. Compare m' with the message from step 2 and explain why perfect secrecy fails.

Then identify the step that uses correctness and the step that uses perfect secrecy.

- (b) **(8 pts)** Apply the bound.
- (i) A prompt is at most 4096 tokens long, each token from a vocabulary of 2^{17} tokens, and every such prompt is possible. How many messages are in $\mathcal{M}$? By part (a), how many bits must the key have? Compare with 256 and say whether both of the vendor's claims can hold.
 - (ii) Write the definition of EAV security (indistinguishable encryption in the presence of an eavesdropper). Compared with perfect secrecy it changes two things: which attackers we quantify over, and how much advantage we allow. Name both.
 - (iii) One of those two changes is why the proof in part (a) no longer applies. Which one, and why? Support your answer by describing what the attacker implicit in part (a) has to compute, and how long that takes.

Problem 3. Pseudorandom generators (30 pts)

A server expands a seed into the coin flips used for sampling, so that a customer who supplies the same seed gets the same output. Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}$ be a secure PRG with $\ell(n) > n$. For each modification below, decide whether the result is a secure PRG. If it is, prove it by reduction to the security of G ; if it is not, give an efficient distinguisher and compute its advantage. If the answer depends on a parameter, state the condition.

- (a) **(6 pts)** $G_1(s) = G(s) \oplus 1^{\ell(n)}$.
- (b) **(6 pts)** The server appends the seed to the stream so that a run can be replayed later: $G_2(s) = G(s) \parallel s$, so $G_2 : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)+n}$.
- (c) **(9 pts)** The client sends two independent seeds $s_1, s_2 \in \{0, 1\}^n$ and the server outputs $G_3(s_1 \parallel s_2) = G(s_1) \oplus G(s_2)$, so $G_3 : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{\ell(n)}$. Answer two questions: for which stretch functions ℓ is G_3 a PRG at all, and is its output pseudorandom? Then suppose a bug makes the client send the same seed twice, $s_1 = s_2$. What does the server output, and which step of your proof used the independence of the seeds?

- (d) **(9 pts)** On one deployment the entropy source is misconfigured and only the last $n/2$ bits of the seed are random; the first $n/2$ bits are always zero, so the server effectively computes $G_4(s') = G(0^{n/2} \| s')$ for $s' \in \{0, 1\}^{n/2}$. Show that the security of G does not imply the security of G_4 : from an arbitrary secure PRG G , construct a PRG $\hat{G}$ that is still secure but whose $\hat{G}_4$ can be distinguished from uniform with advantage $1 - 2^{-\ell(n)}$. Prove both claims about $\hat{G}$. Then explain why the argument “the seed space has only $2^{n/2}$ elements, so try them all” does not by itself give an attack in this model.

Problem 4. One key, many messages (25 pts)

A document store splits each document into chunks and encrypts chunk i of every document in counter mode under a single long-term key k , with the counter initialized to i . Let F be a length-preserving PRF with block length n . The API is

$$\text{Enc}_k(i, m) = \langle i, m \oplus (F_k(\langle i, 1 \rangle) \| F_k(\langle i, 2 \rangle) \| \dots) \rangle,$$

where the caller supplies the chunk index i and m is any polynomial number of blocks.

- (a) **(7 pts)** Prove that this scheme is not CPA-secure. Give the adversary $\mathcal{A}$ in the $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n)$ experiment, including its oracle queries, and compute $\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1]$ exactly. The designers justify the scheme by observing that chunk indices are unique within a document, so no pad is used twice; say what is wrong with that. Then explain why the PRF security of F plays no role in your attack.
- (b) **(7 pts)** Suppose the store holds the vendor’s public FAQ, document A , whose plaintext chunks $m_1^A, m_2^A, \dots$ anyone can obtain, alongside a customer’s confidential document B .
- Given the two ciphertexts, what can an adversary who never learns k compute about B ? How much of B is recovered if B has more chunks than A ?
 - Suppose instead that no plaintext is known. Show that the scheme still leaks an equality pattern across documents: state the predicate the adversary can evaluate, and describe in one or two sentences a realistic harm this causes for a document store.
- (c) **(7 pts)** The design is changed to use a uniformly random $\lambda(n)$ -bit nonce per encryption. The designers want to argue that over $q(n)$ encryptions, where q is a positive polynomial, a nonce repeats only with negligible probability.
- Prove the two lemmas this argument needs. Let $\text{negl}_1, \text{negl}_2$ be negligible and let p be a positive polynomial. Show that $\text{negl}_1 + \text{negl}_2$ is negligible and that $p \cdot \text{negl}_1$ is negligible.
 - Bound the probability that some nonce repeats among $q(n)$ independent uniform $\lambda(n)$ -bit nonces. If $\lambda = 96$ is fixed for every deployment, is that bound negligible as a function of n ? State the condition on $\lambda(\cdot)$ that your argument actually needs, and reconcile it with the fact that 96-bit nonces are used in practice.
- (d) **(4 pts)** Give a corrected design in three or four sentences: what is the nonce, what is scoped per document, and which security definition from Weeks 3–4 is the design aiming at? State the definition; you do not need to prove it.